

Prompt Framing Lab Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	prompt-framing-lab

Big idea: A prompt is not just a request; it frames what kind of knowledge output is likely.

One-Minute Teacher Brief

Students compare how different prompts can make the same AI-style task produce different answers and assumptions.

Student Prompt

Inspect Prompt Framing Lab Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Prompt Framing Lab Stimulus A	Student-facing stimulus 1: Neutral prompt: 'Explain whether homework improves learning.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Prompt Framing Lab Stimulus B	Student-facing stimulus 2: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Prompt Framing Lab Reveal / Twist	Student-facing stimulus 3: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A prompt is not just a request; it frames what kind of knowledge output is likely.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Prompt Framing Lab Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Prompt Framing Lab Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Who is responsible for knowledge produced through a prompt?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how a prompt is not just a request; it frames what kind of knowledge output is likely.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Who is responsible for knowledge produced through a prompt?

Setup Checklist

- Use teacher-written mock outputs if live AI access is not appropriate.
- Keep the task constant while changing only the prompt wording.

Example Stimuli or Materials

- Neutral prompt: 'Explain whether homework improves learning.'
- Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'

Resources To Use

- Resource pack: <resources/prompt-framing-lab/index.html>
- Card deck CSV: <resources/prompt-framing-lab/cards.csv>
- Visual prompt: <resources/prompt-framing-lab/visual-prompt.svg>
- Prompt comparison grid: assumption, evidence requested, uncertainty shown, missing perspective.
- Prompt revision checklist: define task, request sources, ask for uncertainty, include counterarguments.

Run Sequence

1. Give groups the same knowledge task with different prompt wordings.
2. Students predict how their prompt might shape the answer.
3. Groups compare generated or mock outputs for tone, evidence, uncertainty, and omissions.
4. Students revise one prompt to improve reliability and transparency.
5. Debrief how human framing and automated generation interact.

Debrief Moves

- Knowledge question: Who is responsible for knowledge produced through a prompt?
- Can a better question make a machine answer more reliable?

- When does a prompt smuggle in an assumption?

Teacher Quick Script

- The output begins before the machine answers; it begins with the frame we give it.
- What did your prompt make easy for the system to say?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...